From Parents' Cradle to Children's Career: Intergenerational Effects and Complementarities of Childhood Advantage

Sander de Vries Nadine Ketel Maarten Lindeboom*

March 16, 2026

Abstract

Little is known about how much of childhood advantage persists across generations and through which mechanisms. We use birth order as a proxy for childhood advantage and study its intergenerational transmission using Dutch administrative data. A unique feature of our setting is that both generations are 'treated' with the same treatment, allowing us to test whether a repeated "firstborn" advantage produces intergenerational complementarities. We find that 18 percent of the income advantages are transmitted, with no evidence of complementarities across generations. Novel mediation analyses show that differences in partner choice, fertility, and neighborhoods explain nearly half of the transmitted advantage.

Keywords: intergenerational mobility, birth order, extended family, education, crime

JEL Codes: J62, D19, I24, J13

*De Vries: Department of Economics, Vrije Universiteit Amsterdam, s.de.vries@vu.nl, Ketel: Department of Economics, Vrije Universiteit Amsterdam, CEPR, IZA, and Tinbergen Institute, n.ketel@vu.nl; Lindeboom: Department of Economics, Vrije Universiteit Amsterdam, Centre for Health Economics, Monash University, IZA, and Tinbergen Institute, m.lindeboom@vu.nl. We gratefully acknowledge valuable comments from conference and seminar participants in Amsterdam, The Hague, Belgrade, Paris, Potsdam, Melbourne, Essen, Padua, and Copenhagen. The non-public micro data used in this paper are available via remote access to the microdata services of Statistics Netherlands (project agreement 8674). Under certain conditions, these microdata are accessible for statistical and scientific research. For further information: microdata@cbs.nl.

1 Introduction

The intergenerational persistence of income is widely regarded as unfair and a justification for policy intervention (Alesina et al. (2018)). However, we know little about what sustains this cycle or how to break it. One way to make progress is by studying which naturally occurring childhood experiences generate income advantages that persist across generations. Such evidence can clarify the mechanisms through which advantage is transmitted and can help policymakers design interventions that generate long-lasting change.

Treatments in one generation may even increase how much the next generation benefits from *new* treatments. Becker et al. (2018) develop a theory of intergenerational mobility in which human capital increases productivity not only in the labor market, but also in the production of children's human capital. As a result, investments in one generation amplify the returns to parental investments in the next generation. Consistent with this theory, Dettmer et al. (2026) show that Rhesus monkeys benefit only from maternal presence when their mother is also maternally reared. Whether similar intergenerational complementarities exist for humans is an open question.

We break new ground by studying the intergenerational income persistence and complementarity of birth order effects. Prior research shows that children with a higher birth order have unfavorable outcomes across a range of domains, including education, income, and personality traits (Black et al. (2005b); Kantarevic and Mechoulan (2006); Black et al. (2011); Black et al. (2018); Breining et al. (2020)), and that these effects arise early in life (Houmark (2023)).[1] Birth order effects come from sibling comparisons and can thus not be attributed to factors shared by siblings such as schools and neighborhoods. The prevailing hypothesis is that they stem from differences in parental investments. This is supported by compelling evidence that earlier-born children receive more time and resources - especially

[1] In terms of magnitude, birth order effects on education are similar to the effects of compulsory schooling laws often examined to estimate the intergenerational effect of education (Black et al. (2005a), Oreopoulos et al. (2006), Holmlund et al. (2011)). They exceed the effects on college enrollment of a 1 SD higher value-added teacher for 5 years, a 33 percent smaller kindergarten class size for 2 years, or moving from a high-poverty neighborhood (Chetty et al. (2014), Chetty et al. (2011), Chetty et al. (2016)).

during formative early childhood years - and face higher levels of parental stringency (Price (2008), Averett et al. (2011), Hotz and Pantano (2015), Pavan (2016), Black et al. (2018), Lehmann et al. (2018), Lin et al. (2020), Black et al. (2026)).

A unique feature of our setting is that birth order is repeatedly 'assigned' across generations. This allows us to test whether the benefits of parental investments increase if the previous generation also received higher parental investments. For example, firstborns may have better parenting skills through improved education or more stable personality traits. These improved skills could be especially beneficial for their firstborn children because they spend more time with them. This setting resembles Dettmer et al. (2026), who study two generations of randomly induced changes in parenting among Rhesus monkeys, with the key distinction that we study humans.

Following the literature, we use sibling comparisons to estimate birth order effects on the first generation. The main identifying assumption underlying this approach is that parents do not consider the quality of their existing children when deciding to have another child. This assumption is supported by Domingue et al. (2015), Muslimova et al. (Forthcoming), and Isungset et al. (2022), who show that children of different birth orders do not systematically differ in their genetic predisposition for education. To estimate *intergenerational* birth order effects we follow the same framework, but replace the outcomes of siblings with those of their children. We combine these two approaches to test for intergenerational complementarities. Because birth order effects apply to everyone with at least one non-twin sibling and our data cover long-run income for over three million Dutch individuals *and* their children, we can study transmission in a setting that is exceptionally representative and well-powered.

Another contribution of this paper is to study the role of partner choice, fertility, and neighborhood quality in driving intergenerational effects. We develop a framework that makes explicit the biases that arise when such endogenous mediators are naively treated as controls. This framework motivates two new methods that eliminate these biases and allow us to quantify the role of changes in partner choice and the number or timing of children.

These methods can be applied more generally to decompose intergenerational causal effects in other settings.

Our main finding is that around 18 percent of the long-run income advantage due to a low birth order is transmitted to the next generation. For example, the income of firstborn parents is 3 ranks higher than that of thirdborn parents, and their children's income is 0.5 ranks higher than that of children from thirdborn parents. This transmission rate is similar across sex of parents and children. A lower parental birth order also increases educational attainment and reduces boys' criminal behavior, particularly violent crime, consistent with improvements in both cognitive and noncognitive skills.

We find no evidence of intergenerational complementarities. Children benefit from a lower own birth order and a lower parental birth order, but these effects do not interact. The interaction terms are precisely estimated and allow us to rule out complementarities larger than 3 percent of the direct birth order effects on income and education. This suggests that firstborn children do not benefit disproportionately from the improved human capital of their firstborn parents. We conclude that intergenerational complementarities are therefore either limited, context-specific, or masked by offsetting responses in parental investments.

Lastly, using our novel mediation analyses we find that nearly half of the intergenerational effects arise because firstborns match with higher quality partners, move to better neighborhoods, have fewer children, and have them later. Improvements along these dimensions for treated individuals may come at the expense of others because the pool of partners and houses are largely fixed (Abrahamsson et al. (2025)). Based on these results, we argue that the net intergenerational benefits of interventions such as extended education or preschool attendance are potentially overstated if they are mediated by similar channels.[2]

Our results relate to a growing literature on the intergenerational consequences of childhood experiences. Earlier studies mainly focus on changes in parents' education (Currie and

[2]In addition, changes in the number or timing of children affect outcomes of the second generation through birth order effects and cohort-specific trends. We argue in Section 6 that these mediating effects should also be excluded when estimating the net intergenerational benefits of a treatment.

Moretti (2003), Black et al. (2005a), Oreopoulos et al. (2006)), while a more recent literature includes the effects of other childhood experiences like prenatal exposure to pollution, access to health insurance before birth, increased tuberculosis testing and vaccination during school, preschool enrollment, and parental access to disability insurance during childhood (Black et al. (2019); Bütikofer and Salvanes (2020); East et al. (2023); Rossin-Slater and Wüst (2020); Barr and Gibbs (2022); García et al. (2023); Dahl and Gielen (2024)).[3] Particularly closely related are Barclay et al. (2021) and Havari and Savegnago (2022), who already established that a higher parental birth order reduces education.

Our contribution to this literature is threefold. First, we accurately quantify the intergenerational transmission of *income* disadvantages due to a childhood shock. Most existing studies examine earlier outcomes of the second generation, making it difficult to assess the extent to which these childhood experiences break the cycle of long-run income persistence. Second, we provide the first test of intergenerational complementarities of childhood advantage for humans.[4] This enables us to test a crucial assumption in Becker et al. (2018) and can inform policymakers about potentially compounding returns when treatments are sustained across generations. Third, we develop new methods to assess the mediating roles of partner choice and fertility. This improves our understanding of how much of the intergenerational effect is driven by equilibrium reallocation as opposed to improvements in the skills or resources of directly affected parents. Together, these innovations move the literature beyond establishing the existence of intergenerational effects toward understanding their magnitude, mechanisms, and complementarities.

[3]For other papers studying education see Maurin and McNally (2008), McCrary and Royer (2011), Pekkarinen et al. (2009), Holmlund et al. (2011), De Haan (2011), Carneiro et al. (2013), Chevalier et al. (2013), Lundborg et al. (2014), Sikhova (2023), Barrios Fernández et al. (2024), and Akgündüz et al. (2024). The discussion above focuses on Western countries. There are a few studies in non-western countries, such as the intergenerational effects of school construction (Mazumder et al. (2023), Akresh et al. (2023b)), deworming (Walker et al. (2023)), and war exposure (Akresh et al. (2023a)).

[4]This is also related to a growing number of papers that study complementarities of *different* treatments within the *same* generation. See Almond et al. (2018) for a review.

2 Data

Sample. We use administrative data from Statistics Netherlands covering the full population and select all individuals born in the Netherlands between 1945 and 1970, dropping all children in families of migrants (5.7%), with at least one missing birth date (8.4%), twins (5.7%), only one child (8.8%) or six or more children (17%).[5] We refer to this sample as the first generation; this includes 64 percent of all individuals born between 1945 and 1970. To establish birth order we rank individuals with the same mother and father by their birth dates, so our analysis focuses solely on birth order among full siblings. The first generation is linked with their children, referred to as the second generation.[6] In the core analysis, we focus on children born before 1991.

Parent and child income. The income register records the gross personal income extracted from tax statements spanning the period between 2003 and 2023, and includes all income from employment, entrepreneurship, income insurance payments, and social security benefits. We measure income in 2024 euros, adjusting for inflation using the consumer price index. Household income is defined as the total income of all household members.[7] Household income provides a reliable measure of economic resources even in the case of non-participation in the labor market and is commonly used in other intergenerational mobility studies (Chadwick and Solon (2002)). We observe household income records for 97.3% of the second generation sample and 97.1% of the first generation sample.

A well-known challenge is that income snapshots are prone to measurement error due to transitory income shocks (Mazumder (2005)) and life-cycle bias arising from heterogeneous

[5] We drop migrants because family links are incompletely observed and because these families often arrive simultaneously, creating a correlation between birth order and age at arrival of siblings. Single-child families are not used for identification; we drop larger families for conciseness of our results.

[6] We rely on legal relationships between parents and children. Consequently, the identified parent is not necessarily the biological parent, but rather one who has most likely raised the children. Parents do not need to be together for the (full) period in which they raise their children.

[7] If children still live with their parents, the parents' household income is defined as the joint parental income, and the child's income as their personal income.

age-income profiles (Haider and Solon (2006)). Since income is only used as an outcome, attenuation bias due to transitory shocks is no concern. Nevertheless, we still average incomes over four years for precision. To mitigate life-cycle bias, we restrict these observations to the four years closest to age 35, and within the age range of 30 to 60 (Nybom and Stuhler (2017)).[8] We define individuals' income ranks based on their positions in the distribution of long-run household income in their respective cohorts. To assess the robustness of our findings, Section 4 reports results using alternative income measures and age windows.

Children's education and crime. We construct an indicator for a higher education degree based on the education register that contains tertiary education degrees since 1986. The primary crime outcome is an indicator of whether a child has been suspected of any crime at ages 18 to 20, which are the prime ages at which individuals commit crimes in the Netherlands. The analysis on crime is restricted to boys and distinguishes between property crime, violent crime, and other types of crime based on the reported offense types. The crime outcomes are available for a shorter period, so the crime sample includes only children born between 1986 and 2001.

Descriptives. Table 1 presents descriptive statistics for the parents and the children separated by the parent's birth order. Panel A shows descriptives for the first-generation sample. Panel B includes all their children born before 1991, which are the children we will use for the income and education analyses.[9] Many children in the analysis sample occur once for each parent, and thus twice in the dataset.[10]

Consistent with the earlier literature on birth order effects, first-generation individuals with a higher birth order on average have a lower income (Black et al. (2005b)). From the

[8] We observe parental income around ages 45 to 60, and for children around ages 30 to 45. Only 1.6 percent of the children and 2.6 percent of the parents have less than four income observations.

[9] For the crime analysis we rely on a different sample of children born between 1986 and 2001. Sample averages for crime can be found in Table A6.

[10] We explain this in greater detail in Section 3.

children's outcomes in panel B we observe that children of higher-birth-order parents have lower income and less often complete higher education. However, these are just correlations. We next explain how we identify causal (parental) birth order effects.

Table 1: Descriptive Statistics

A. First Generation	Birth Order				
	1	2	3	4	5
Year of birth	1959.1	1959.1	1958.9	1958.9	1959.2
Male	51.3	51.2	51.1	51.2	51.0
Household income rank	55.2	53.5	52.5	51.5	50.6
Has child	79.0	78.2	78.3	78.2	77.6
Age at first child	29.0	28.9	28.7	28.5	28.4
Number of children	1.7	1.7	1.7	1.8	1.7
N	1,225,906	1,136,740	560,688	240,050	78,603
B. Second Generation	Parental Birth Order				
	1	2	3	4	5
Year of birth	1981.6	1981.5	1981.5	1981.7	1981.9
Male	51.0	50.9	50.9	51.0	50.9
Household income rank	55.1	54.9	54.9	54.9	54.7
Higher education completion	41.6	41.0	40.6	40.1	40.2
N	1,077,176	985,835	511,243	225,344	71,897

Notes: The sample in panel A includes all individuals born between 1945 and 1970 who meet the sample selection criteria outlined in Section 2. Panel B includes all children of individuals from panel A who were born before 1991. Outcomes are categorized by the birth order of the parent. Some children are in the sample twice: once for their father's birth order and once for their mother's birth order. The fertility outcomes in panel A include all children, including those born after 1991. The average income rank differs from 50 because income ranks are calculated with the full sample before excluding singletons, migrants and twins. All cells represent sample averages.

3 Identification

First generation effects. Our identification strategy relies on within-family variation in birth order. To estimate birth order effects for the first generation, we estimate the following

Sibling Fixed Effects (SFE) model

$$Y_{pf} = \alpha_f + \sum_{k=2}^{5} \beta_k^{FG} I[BO_{pf} = k] + \tau_{t(pf)} + \epsilon_{pf}, \tag{1}$$

where Y_{pf} is the outcome of a child p in a family f, α_f are family fixed effects, $I[BO_{pf} = k]$ is an indicator that equals 1 if the birth order of child p equals k, and $\tau_{t(pf)}$ are year of birth $\times$ month of birth $\times$ sex $\times$ family size fixed effects. The family fixed effects ensure that we only compare siblings. By including $\tau_{t(pf)}$, we flexibly control for different trends in the outcome between cohorts by gender and family size. We model the fixed effects by year and month to ensure that even when two siblings are born in the same year (but are not twins), their difference in birth timing is controlled for. The coefficients β_k^{FG} capture the birth order effects on the first generation.

Although the family fixed effects rule out confounders that differ between families, there can still be within-family confounders. In particular, birth order effects can arise mechanically if parents' fertility decisions are related to the quality of their earlier children. When parents stop having children after having a particularly 'bad draw', then birth order effects are the result of the last child being negatively selected. Domingue et al. (2015), Muslimova et al. (Forthcoming), and Isungset et al. (2022) find that children of different birth orders do not structurally differ in their polygenic score for education. This suggests that, at least genetically, children of different birth orders are of similar 'quality'.[11] Moreover, in our main results, we also compare first and second-born children in families of three or more, who should not be impacted by such an optimal stopping rule. Our results are virtually unchanged for these comparisons.

By construction, birth order effects also capture the effect of having older or younger siblings. For example, in a family of size two, the effect of being born second includes the

[11] Generally, it can be hard to infer an earlier child's 'quality' at the time that parents decide about another child. One of the few signals parents have at this early stage is children's health. However, unlike most outcomes later in life, firstborn children tend to have worse health outcomes relative to later-born siblings during their first years (Brenøe and Molitor (2018)). This contradicts the idea of an optimal stopping rule.

effect of having an older sibling. As a result, birth order effects may arise from spillovers between siblings. However, contrary to birth order patterns, most studies find that if there are spillovers, they make siblings more similar (e.g. Dahl et al. (2014), Nicoletti et al. (2018), Bharadwaj et al. (2022)). Moreover, multiple papers show that having a sibling or having more siblings does not affect children's outcomes (Black et al. (2005b), Angrist et al. (2010), Ilciukas et al. (2025)). This suggests that sibling spillovers are unlikely to be a major driver of birth order effects.

Intergenerational effects. To estimate our effect of interest, the intergenerational effect of birth order, we replace the outcomes of children p with the outcomes of their children, indexed by cp. This results in the following Cousin Fixed Effects (CFE) model

$$Y_{cpf} = \alpha_f + \sum_{k=2}^{5} \beta_k^{SG} I[BO_{pf} = k] + \tau_{t(pf)} + \epsilon_{cpf}, \tag{2}$$

where Y_{cpf} is the outcome of child c of parent p in extended family f, α_f are extended family fixed effects, $I[BO_{pf} = k]$ is an indicator that equals 1 if the birth order of parent p equals k, and $\tau_{t(pf)}$ are parent's year of birth $\times$ month of birth $\times$ sex $\times$ family size fixed effects. The extended family fixed effects ensure that we only compare *the children* of siblings. These children are cousins, but only from one side of the family. The regressions are weighted by the number of children so that all parents receive equal weight.

In our analysis, we compare children once to cousins from their mother's side and once to cousins from their father's side. Thus, most children occur twice in the dataset.[12] These results are averaged into a single treatment effect that captures both paternal and maternal birth order effects. We also explore heterogeneity in the effects by studying the effects for fathers and mothers separately.

[12]This would be problematic when children are compared to themselves. This is unlikely in our design because this only happens when a brother and sister have children together. Some children are sampled only once because one of the parents does not meet the sample selection criteria.

Intergenerational complementarity. We next want to test whether birth order effects interact across generations. Identifying this interaction requires independent variation in parental birth order and child birth order. A challenge is that parental birth order effects are identified using cousin comparisons, but among cousins, differences in birth order can be correlated with unobserved between-household differences, such as family size or parental income.

To address this, we proceed in two steps. We first isolate within-household variation in child birth order. Specifically, we regress child birth order on sibling and cohort fixed effects:

$$BO_{cpf} = \alpha_{pf} + \tau_{t(cpf)} + \nu_{cpf}$$

The residual of this regression, $\tilde{BO}_{cpf} = BO_{cpf} - \hat{BO}_{cpf} = BO_{cpf} - \hat{\alpha}_{pf} - \hat{\tau}_{t(cpf)}$, measures precisely the within-household variation in birth order we use to estimate birth order effects (as in equation 1).

In the second step, we re-estimate the cousin fixed effects model (equation 2), now including the residualized birth order and its interaction with parental birth order:

$$Y_{cpf} = \alpha_f + \beta^{FG} BO_{pf} + \beta^{SG} \tilde{BO}_{cpf} + \beta^{SG \times FG} BO_{pf} \times \tilde{BO}_{cpf} + \tau_{t(pf)} + \epsilon_{cpf}. \tag{3}$$

Intuitively, this specification tests whether birth order effects differ between children of later-born siblings and children of earlier-born siblings. Because equation 3 relies only on within-household variation in child birth order, and parental birth order is fixed within households, $\tilde{BO}_{cpf}$ is by construction independent of BO_{pf}. The coefficient $\beta^{SG \times FG}$ therefore cleanly identifies the interaction between parental and child birth order. Standard errors are computed using a clustered bootstrap to account for the two-step estimation procedure.

4 Intergenerational Transmission

We begin by estimating the effects of birth order on the income ranks of the first and second generation. Table 2 displays the estimates. Column 1 includes the full analysis samples; columns 2 to 5 present (parental) birth order effects separately by the first generation's family size.

Panel A shows that parental disadvantages due to their birth order transmit to their children. For example, column 1 shows that the income rank of children of a thirdborn parent is 0.5 ranks lower than their cousins of a firstborn parent. The effects increase with each additional birth order of the parent. Columns 2 through 5 show consistent patterns across different extended family sizes. These results highlight that birth order has considerable intergenerational effects.

While the effects in panel A are interesting on their own, they are not informative about *how much* of parental disadvantage is transmitted to their children. To shed light on this, we present the first-generation birth order effects on income in panel B. In line with Black et al. (2005b), these estimates show that birth order has large effects on income.[13] We also observe that the effects of being second or thirdborn are similar for all family sizes. This is inconsistent with optimal stopping models of fertility, which would suggest that only the last-born has particularly low income.

We use the estimates in panel B to obtain the degree of intergenerational transmission, computed as the ratio of intergenerational birth order effects (β^{SG}) to birth order effects on the first generation (β^{FG}). A transmission coefficient above (below) one indicates that the intergenerational birth order effects are greater (smaller) than those on the first generation. This ratio is reported in panel C. The estimates in column 1 are around 0.18 and columns 2 to 4 show that this transmission rate is relatively consistent across families of different sizes and various parental birth orders. Aggregating these estimates into one coefficient using

[13] The sample in panel B includes all individuals from the first generation, including those without children born before 1991. Using the sample of parents with children in our core sample yields very similar estimates (Table A1).

2SLS yields a transmission estimate equal to 0.18 (standard error 0.04). We conclude that on average about 20 percent of the income disadvantages due to a higher birth order are transmitted to the next generation.

For comparison, the rank-rank correlation between parental and child income is 0.25 in our sample. This implies that the persistence of income disadvantages due to birth order is slightly weaker than the overall persistence of income across generations.

Whether the transmission of income advantages due to birth order effects of 0.2 is considered high or low depends on one's priors. On the one hand, treatments that affect parents' education or income do not always result in intergenerational effects, or result in considerably smaller effects than conventional intergenerational mobility estimates suggest (Holmlund et al. (2011), Page (Forthcoming)). In this light, the intergenerational effects of birth order are sizable. On the other hand, the treatments that parents are exposed to in these studies result in changes in education or income at a relatively old age, while birth order affects both cognitive and noncognitive skills from young ages onward (Houmark (2023)). These initial disadvantages might compound over time and result in relatively large effects for subsequent generations (Becker et al. (2018)). An extreme example comes from Barr and Gibbs (2022), who show that the intergenerational effects of attending preschool education even surpassed the initial impact on the parents. From this perspective, the intergenerational effects of birth order appear relatively modest.

Gender differences. We test for gender differences in Table 3. For readability, we have converted birth order into a numeric variable from 1 to 5. The results do not differ much if we impose this restriction and it makes the estimates more precise. Panel A shows that a higher birth order decreases household income slightly more for males. Panel B column 1 shows that the intergenerational effects are also somewhat (but not significantly) larger for males, and these effects do not differ much depending on the gender of the child. Column 2 shows that the ratio of second-generation to first-generation effects — the intergenerational

Table 2: Birth Order Effects on the First and Second Generation's Income Ranks

Birth Order	All (1)	Family size 2 (2)	Family size 3 (3)	Family size 4 (4)	Family size 5 (5)
	A. Intergenerational Birth Order Effects (β^{SG})				
2	-0.309***	-0.405**	-0.171	-0.424***	-0.465***
	(0.066)	(0.159)	(0.115)	(0.121)	(0.152)
3	-0.538***		-0.627***	-0.485**	-0.507**
	(0.116)		(0.219)	(0.190)	(0.208)
4	-0.712***			-0.662**	-0.741**
	(0.187)			(0.299)	(0.296)
5	-0.831***				-0.840**
	(0.304)				(0.425)
Mean	54.967	54.795	55.088	55.073	54.898
SD	27.208	27.429	27.239	27.099	26.894
N	2,871,495	853,322	881,249	679,974	456,950
	B. Birth Order Effects (β^{FG})				
2	-1.577***	-1.789***	-1.585***	-1.527***	-1.295***
	(0.053)	(0.114)	(0.085)	(0.104)	(0.149)
3	-2.958***		-3.060***	-2.788***	-2.789***
	(0.094)		(0.157)	(0.151)	(0.185)
4	-4.106***			-3.972***	-3.744***
	(0.155)			(0.233)	(0.250)
5	-5.162***				-4.734***
	(0.258)				(0.351)
Mean	53.766	54.607	54.299	52.835	51.203
SD	27.901	28.053	27.925	27.726	27.445
N	3,241,987	1,192,284	1,032,476	644,990	372,237
	C. Degree of Intergenerational Transmission (β^{SG}/β^{FG})				
2	0.196***	0.226**	0.108	0.278***	0.359***
	(0.042)	(0.091)	(0.073)	(0.082)	(0.123)
3	0.182***		0.205***	0.174**	0.182**
	(0.040)		(0.073)	(0.069)	(0.076)
4	0.173***			0.167**	0.198**
	(0.046)			(0.076)	(0.08)
5	0.161***				0.177*
	(0.060)				(0.091)

Notes: Panel A reports parental birth order effects, estimated according to equation 2. Panel B reports birth order effects on the first generation, estimated according to equation 1. Panel C reports the ratio of the estimates in panel A to the estimates in panel B. These ratios are computed using two-sample 2SLS. Standard errors are in parentheses. Standard errors in panel A are clustered on the household level. Standard errors in panel C are based on the two-sample 2SLS correction from Inoue and Solon (2010). (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

transmission rate — does not differ substantially by parent gender. Overall, these results suggest that birth order effects are transmitted similarly across genders.

Robustness. Table A2 shows that the degree of intergenerational transmission is very similar when we use the log of household income or personal income ranks. Table A3 shows that the estimates remain stable when varying the number of income observations. Figure A1 shows that first-generation birth order effects are similar when income is measured at any age between 33 and 60. The effects are slightly attenuated at ages 30 to 32, suggesting that income measured this early may not accurately reflect long-run income. In our baseline specification, these younger income observations are used only for the most recent cohorts (comprising 20 percent of the sample). Reassuringly, the results hardly change when we exclusively use income observations above age 32 (Table A3).

Education and Crime. We next estimate parental birth order effects on education and crime. We do not have information on the parents' university enrollment or criminal activity at the same ages as their children, thus precluding a direct calculation of intergenerational transmission. Instead, we compare the parental birth order effects to the second generation's birth order effects. For completeness, we also report results for income.

Figure 1 shows that parental birth order significantly decreases children's higher education attainment. For example, children of thirdborn parents are 1.6 percentage points (four percent) less likely to have a higher education degree than children of firstborn parents. A higher parental birth order also increases boys' likelihood to be suspected of a crime.[14] The ratio of parental birth order effects to birth order effects is roughly between 20 and 30 percent for all three outcomes.

Even though the parental birth order effects are smaller than birth order effects, their magnitude is non-trivial. For example, parental birth order increases boys' crime by up to

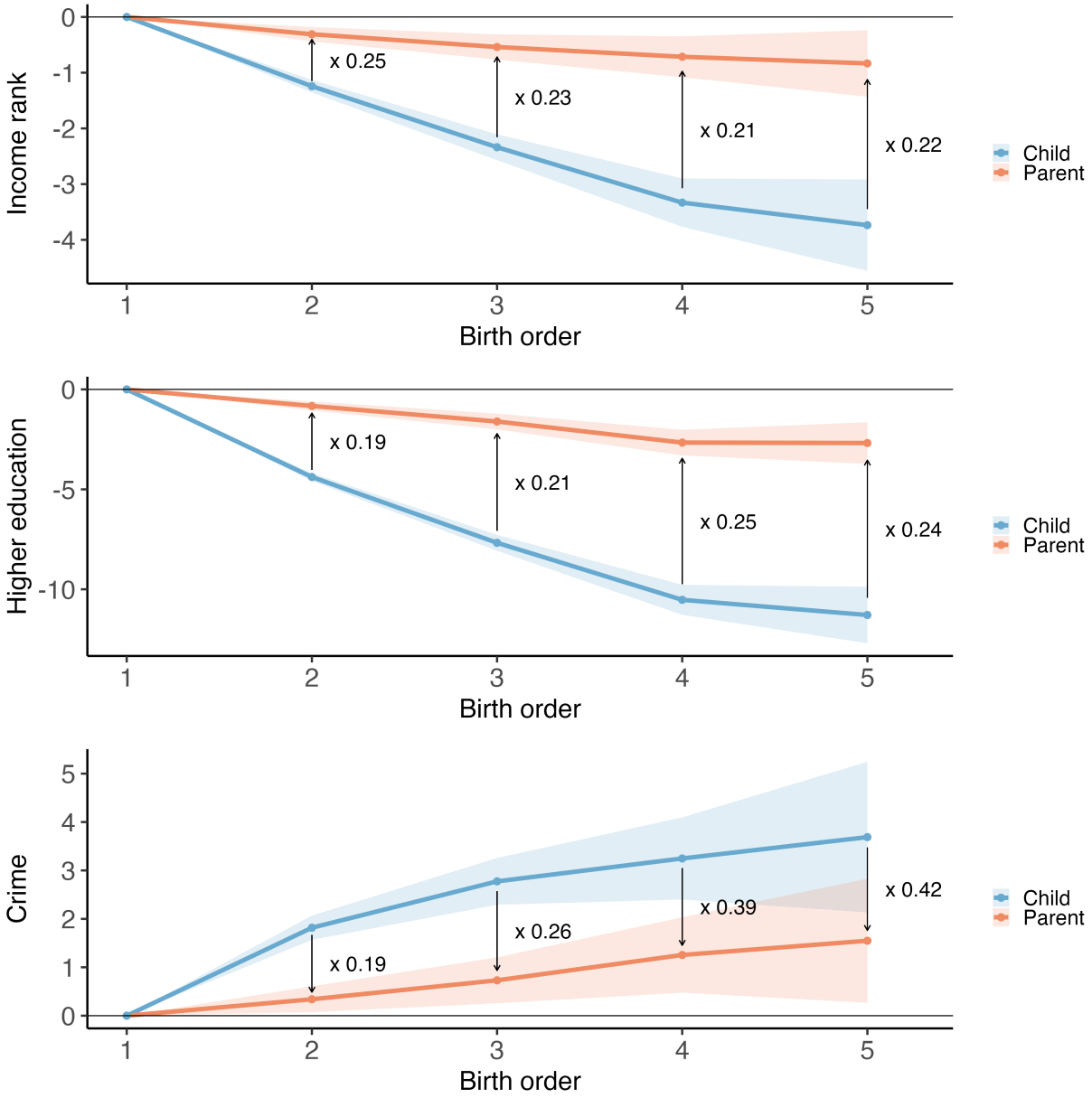
[14]Estimates by family size are reported in Tables A4 and A5.

Table 3: Gender Differences

A. Birth order effects on the first generation	Household income rank (β^{FG})	
Birth order × female	-1.393***	
	(0.050)	
Birth order × male	-1.537***	
	(0.049)	
N	3,241,987	
B. Intergenerational birth order effects	Household income rank child (β^{SG})	Degree of transmission (β^{SG}/β^{FG})
Maternal birth order effect on daughter	-0.210***	0.151***
	(0.068)	(0.049)
Maternal birth order effect on son	-0.222***	0.159***
	(0.067)	(0.048)
Paternal birth order effect on daughter	-0.363***	0.236***
	(0.070)	(0.047)
Paternal birth order effect on son	-0.291***	0.190***
	(0.069)	(0.046)
	2,871,495	2,871,495

Notes: Panel A reports birth order effects on household income, estimated according to 1, where birth order is interacted with gender. Panel B column (1) reports *parental* birth order effects, estimated according to 2, where parental birth order and the cohort fixed effects are interacted with the gender of the child. Panel B column (2) reports the ratio of the estimates in column (1) to the estimates in panel A, with standard errors based on the two-sample 2SLS correction from Inoue and Solon (2010). Standard errors in panel B column 1 are clustered on the household level. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Figure 1: Parental Birth Order Effects, Birth Order Effects, and their Ratio



Notes: The blue coefficients represent birth order effects, estimated using equation 1, while the orange coefficients capture parental birth order effects, estimated using equation 2. The first panel presents results for household income ranks, based on 2,865,052 observations. This sample includes all children from Table 2, excluding 6,443 children with missing birth order. The second panel presents results for higher education completion, based on 2,952,774 observations. This sample includes all children who meet the sample selection criteria in Section 2, including those with missing income. The third panel examines whether a son is suspected of a crime between ages 18 and 20, with a sample size of 1,515,074. The estimates of parental birth order effects for education and crime can also be found in Table A4 and A5, respectively. The arrows represent the ratio of parental birth order effects to individual birth order effects. Shaded areas denote 95 percent confidence intervals.

10 percent for thirdborn children (relative to the sample mean). In Table A6 we show that the rise in crime is primarily driven by violent offenses. In this category, effect sizes are as high as 20 percent.

We conclude that birth order has important economic consequences that transmit across generations. We next test whether it also affects how the second generation responds to its own birth order.

5 Intergenerational complementarity

In the presence of dynamic complementarities, early investments raise the returns to later investments (Cunha and Heckman (2007)). Applied across generations, this logic implies that improved human capital of one generation raises the productivity of parental investments in the next generation (Becker et al. (2018)).

Dettmer et al. (2026) test for such intergenerational complementarities with Rhesus monkeys. They study two generations of monkeys who were randomly assigned to be raised by either their mother or a human caretaker. The results show that monkeys benefit from being raised by their mother, but only if their mother was also maternally reared. This illustrates that early-life advantage in one generation can amplify the benefits of parental investments into the next. Identifying intergenerational complementarities for humans is challenging because it requires a multigenerational experiment in which the first and second generation are independently and randomly assigned to receive different levels of parental investments.

Our setting closely resembles such an experiment. Birth order acts as a treatment that, like the treatment in Dettmer et al. (2026), generates different levels of parental investments and is plausibly randomly assigned across two generations. As with maternally reared monkeys, individuals with a low birth order experience early-life advantages that lead to large improvements in human capital, including education, income, noncognitive skills, and personality traits (Black et al. (2005b), Black et al. (2018), Breining et al. (2020)). These

advantages may improve the quality of parental investments they later provide. Such improvements may be especially valuable for their firstborn children because they receive more parental investments. This amplification of birth order effects would be consistent with the existence of intergenerational complementarities.

Results. We exploit two generations of within-family variation in birth order to test for intergenerational complementarities (equation 3). The results are reported in Table 4. Consistent with our prior findings, we see that children benefit from a lower birth order and a lower parental birth order, and the magnitude of the intergenerational effect is 20 to 30 percent of the direct birth order effects across all three outcomes. A significant interaction term with the same sign as the birth order effect would be consistent with intergenerational complementarities. We find the opposite. None of the interaction terms are significant and for income and education we rule out complementarities larger than 3 percent with 95 percent confidence. Because mothers are often the primary caregiver, the interaction effect could be stronger for mothers. However, even for mothers the interaction terms are precisely estimated at zero (Table A7). We conclude that firstborn children do not benefit disproportionately from their parents' improved human capital.

Table 4: Intergenerational Complementarity

	(1) Income	(2) Education	(3) Crime
Birth order (β^{FG})	-1.296***	-4.145***	1.335***
	(0.116)	(0.189)	(0.256)
Parental birth order (β^{SG})	-0.277***	-0.830***	0.322***
	(0.049)	(0.092)	(0.108)
Birth order × parental birth order ($\beta^{FG \times SG}$)	0.049	0.053	0.056
	(0.048)	(0.085)	(0.111)
Complementarity effect size ($\beta^{FG \times SG}/\beta^{FG}$)	-3.8%	-1.3%	4.2%
	[-10.5%, 2.9%]	[-5.2%, 2.6%]	[-16.0%, 24.4%]
N	2,871,495	2,871,495	1,521,986

Notes: Each column reports birth order effects, parental birth order effects, and their interaction effect estimated according to equation 3. Column 1 presents results for household income ranks. Column 2 presents results for higher education completion. The outcome in column 3 is an indicator for whether a son is suspected of a crime between ages 18 and 20. The sample in Column 3 includes only sons. Standard errors are in parentheses. The standard errors are based on a clustered bootstrap with 200 replications. The bottom row reports the 95 percent confidence intervals for the complementarity effect size estimates in brackets. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Discussion. We discuss three potential explanations for why we do not find interaction effects. First, higher parental human capital need not necessarily translate into better parenting skills. If a low birth order improves adult outcomes without increasing the efficiency with which parents convert inputs into child human capital, then complementarities will not arise. This would be inconsistent with the theory in Becker et al. (2018).

Second, the improvements in parental human capital due to a low birth order may be too small to generate meaningful complementarities. This explanation is not promising for policy. If human capital improvements due to birth order do not produce complementarities, then most interventions are unlikely to generate them either because they often result in smaller improvements. For example, the effect on higher education completion of being firstborn versus thirdborn (+7.5 p.p.) is considerably larger than the effect on college enrollment of a 1 SD higher value-added teacher from grades 4 to 8 (+4 p.p., Chetty et al. (2014)), a 33% smaller kindergarten class size for two years (+1.8 p.p., Chetty et al. (2011)), or moving

from a high-poverty neighborhood before age 13 (+5 p.p., Chetty et al. (2016)).[15]

Third, birth order could affect both the quality and quantity of parental investments, with offsetting effects. For example, improved human capital may raise the productivity of parental investments and simultaneously reduce time spent with children through higher labor supply or greater use of outsourced childcare. In this case, intergenerational complementarities could exist but cancel out in observed outcomes.

We view our study as a starting point for a broader research agenda on intergenerational complementarities among humans. Our results suggest that complementarities are either limited, context-specific, or masked by offsetting responses in parental investments. Future work using alternative sources of improvements in parental human capital and richer measures of parental investments is needed to differentiate between these explanations.

6 Partner Choice, Fertility, and Neighborhood Quality

In Section 4 we show that 18 percent of income advantages due to a lower birth order persist into the next generation. Because many years elapse between parents' early-life advantages and the outcomes of their children, these intergenerational effects could arise through several channels. There are four main channels that we can think of: (i) the directly affected parent, (ii) the partner, (iii) the neighborhood, and (iv) fertility.

Distinguishing between these channels is not only important for understanding the mechanisms that drive intergenerational persistence, but also for assessing whether the transmitted advantages reflect net social gains. Persistence driven by improved skills and resources of directly affected parents represents a net gain, whereas persistence operating through partner choice or neighborhood quality may not. The pool of partners and the housing stock are largely fixed, so improvements for treated individuals can come at the expense of others (Abrahamsson et al. (2025)). Ignoring such general equilibrium effects then leads to an

[15]The papers above only measure effects on college enrollment rather than completion. Since completion is a stricter margin than enrollment, our effect sizes likely understate the comparison relative to those papers.

overstatement of the net intergenerational benefits of a treatment.

Changes in fertility represent another mediating mechanism that does not necessarily imply social gains. When the number or timing of children change, intergenerational outcomes may partly be driven by birth order effects and cohort-specific trends in the second generation. These differences reflect parents spreading resources over fewer children or children being born in different years with different conditions rather than improved parental investments.[16]

This distinction between direct and indirect channels has received limited attention in the literature so far. When addressed, partner quality or child birth order and year are typically added as controls to the treatment specification (Holmlund et al. (2011), Barclay et al. (2021)).[17] For changes in birth order specifically, some studies remove all later-born children (Currie and Moretti (2003), Rossin-Slater and Wüst (2020)). However, changes in fertility or partner choice are endogenous and require additional exogenous variation to isolate their importance. We are also not aware of studies that consider the role of neighborhoods in explaining intergenerational causal effects.[18]

We develop a framework that makes explicit the biases that occur when these mediators are included as (bad) controls. This framework motivates two novel methods that address these biases for partner choice and fertility. These methods are generally applicable to decomposing intergenerational effects. We study the role of neighborhoods by controlling for granular neighborhood fixed effects. Although neighborhood choice is also endogenous, we argue that this analysis provides informative bounds on its contribution.

[16] Additionally, reduced fertility has important consequences for the future workforce.

[17] Several studies examine assortative mating as a determinant of aggregate intergenerational mobility (Fernández and Rogerson (2001), Ermisch et al. (2006), Holmlund (2022), Collado et al. (2023)). Evidence from causal settings is limited. A notable exception is Abrahamsson et al. (2025), who explicitly study assortative mating and show that it likely explains why in their setting intergenerational effects arise for treated mothers but not for fathers.

[18] There is a large literature studying the role of neighborhoods in driving aggregate intergenerational mobility (Chyn and Katz (2021), Mogstad and Torsvik (2023)). However, we are not aware of papers that study their importance in sustaining causal effects across generations.

6.1 Partner choice

The goal is to decompose the total effect of parental birth order into a component explained by partner choice and a remaining unexplained component. Let Y_{cp} denote child income and let BO_p denote birth order of the treated parent p. Let S_{cp} denote the human capital of the treated parent's partner associated with child c. This variable captures all characteristics of the partner which matter for child income. Throughout this section, we omit the control variables (cousin and cohort fixed effects) for notational convenience. All variables and error terms should be interpreted as residualized with respect to the full set of controls.

We consider the following causal model:

$$Y_{cp} = \beta_1 BO_p + \beta_2 S_{cp} + \nu_p + e_{cp}, \tag{4}$$

$$S_{cp} = \delta_1 BO_p + \eta_{pc}. \tag{5}$$

In equation 4, β_1 captures all channels through which parental birth order affects child income other than through the mediator, partner human capital. The coefficient δ_1 measures the effect of birth order on the mediator, while β_2 captures the effect of the mediator on child income. In this framework, the total treatment effect of parental birth order on child income equals $\beta = \beta_1 + \delta_1\beta_2$. The second term captures the indirect effect of parental birth order operating through changes in partner human capital.

Naturally, there is residual variation in child income and partner human capital which is unrelated to birth order. In equation 5, η_{pc} reflects this residual variation in partner human capital S_{cp}. We decompose residual variation in child income Y_{cp} into a component reflecting unobserved characteristics of the treated parent (ν_p) and child-specific shocks (e_{cp}). Partner characteristics η_{cp} are potentially correlated with unobserved characteristics of the treated parent, such that $Cov(\eta_{cp}, \nu_p) \neq 0$. We assume that child-specific shocks are independent of partner quality, so that $Cov(\eta_{cp}, e_{cp}) = 0$. This reflects the idea that children are born after parents match.

Since birth order is by assumption as-good-as-randomly assigned conditional on the controls, a regression of Y_{cp} on BO_p identifies the total effect (β) and a regression of S_{cp} on BO_p identifies the effect on the mediator (δ_1). The parameters β_1 and β_2 are not directly identified.

Using endogenous variation in S_{cp}. A common approach to mediation analysis is to include the endogenous mediator in the treatment effect specification. The coefficient on the treatment variable is then interpreted as the part of the treatment effect that cannot be explained by the mediator.

This approach is problematic in our setting because partner human capital is not exogenous. Intuitively, given that most parents have children with only one partner, comparing children across different levels of partner human capital implies comparing children of different couples. This makes it difficult to disentangle the role of the partner from that of the treated parent.

Formally, including partner human capital S_{cp} in the regression of Y_{cp} on BO_p identifies

$$\beta_2^{\text{Naive}} = \beta_2 + \frac{\text{Cov}(\eta_{pc}, \nu_p)}{\text{Var}(\eta_{pc})}. \tag{6}$$

This expression shows that the coefficient on partner human capital equals the causal effect of partner quality plus selection effects. When unobserved parental traits that raise child income (ν_p) are positively correlated with partner human capital (η_{cp}), as in the case of assortative mating, the naive estimate is biased upward. In that case, too much of the total treatment effect is attributed to differences in partner human capital.

A novel approach. A more credible mediation analysis requires a counterfactual comparison of child income for the same parent across partners with different human capital.

We propose a novel approach to identify β_2 using parents who have children with multiple partners. This setting allows us to estimate how one parent's human capital relates to child income while holding fixed all time-invariant characteristics of the other parent, such as

education, genetic endowments, or stable personality traits.

To formalize this approach, we rewrite equation 4 as

$$\begin{aligned} Y_{cp} &= \underbrace{\beta_1 BO_p + \nu_p}_{\alpha_p} + \beta_2 S_{cp} + e_{cp} \\ &= \alpha_p + \beta_2 S_{cp} + e_{cp}. \end{aligned} \tag{7}$$

This corresponds to a regression of child income on parent fixed effects α_p and partner human capital S_{cp}. Identification of β_2 comes from parents who have children with multiple partners who vary in human capital S_{cp}. All characteristics of the focal parent that do not change across partnerships are absorbed by α_p. This specification delivers a consistent estimate $\hat{\beta}_2^{FE}$ for the relationship between partner human capital and child income.

Once β_2 is identified, identification of β_1 follows by removing the mediating effect of partner human capital from child income:

$$\tilde{Y}_{cp} = Y_{cp} - \hat{\beta}_2^{FE} S_{cp} = \beta_1 BO_p + \xi_{cp}, \tag{8}$$

where $\xi_{cp} = \nu_p + e_{cp} + (\beta_2 - \hat{\beta}_2^{FE}) S_{cp}$. Parental birth order is exogenous by assumption so that $Cov(BO_p, \nu_p) = Cov(BO_p, e_{cp}) = 0$ and in large samples $(\beta_2 - \hat{\beta}_2^{FE}) S_{cp} = 0$. As a result, $\text{Cov}(BO_p, \xi_{cp}) = 0$ and a regression of $\tilde{Y}_{cp}$ on BO_p identifies β_1.

To summarize, we implement the mediation analysis in two steps. First, we estimate the effect of the mediator, partner human capital, on child income. We use the income rank of the partner as a proxy for partner human capital. Second, we remove the estimated mediating component from child income and re-estimate the standard treatment effect specification using the residualized outcome. We compute standard errors using a clustered bootstrap to account for the two-step estimation procedure.

Results. Column 1 in panel A of Table 5 shows that a one unit increase in paternal

birth order reduces maternal income by 0.37 ranks. The corresponding estimate for maternal birth order is 0.47 ranks.[19] These estimates indicate that partner human capital could be an important mediator ($\delta_1 \neq 0$). The key question is how much changes in partner human capital translate into differences in child income.

We address this using our novel design. We select all children with a half-sibling in the core sample (1.7 percent) and estimate the association between partner income and child income in panel B. Among children with maternal half-siblings, a one rank increase in partner (father) income is associated with a 0.26 rank increase in child income. Consistent with upward bias due to assortative mating (equation 6), the coefficient falls substantially when adding mother fixed effects in column 2. The estimate implies that, holding the mother fixed, a one rank increase in father income is associated with a 0.10 rank increase in child income. For maternal income we find a cross-sectional association of 0.19, which reduces to 0.09 when including father fixed effects.[20]

We use these estimates to complete our mediation analysis. Following equation 8 we subtract the implied mediating effect due to different partner choice from child income and estimate a cousin fixed effects model using the residualized outcome (panel A, column 3).[21] The difference between these estimates and the estimates in column 2 reflect the share of the total effect driven by partner choice, as proxied by partner income. The implied shares are calculated in Column 3 in panel B and are 17 percent for mothers and 16 percent for fathers. This shows that changes in partner choice can account for an economically relevant share of the transmitted advantages.

[19] To put this in perspective, a one unit increase in birth order reduces own income by about 1.1 ranks (Table A2).

[20] If later partners are systematically of higher quality (high η_{cp}) and time varying characteristics of focal parents also improve (high e_{cp}), this could bias β_2. Inconsistent with this, the estimates are robust to controlling for the partner's age at birth, the focal parent's age at birth, and relationship order (Table A12). Nonetheless, we cannot rule out time variation in parental characteristics that is correlated with partner quality but not captured by these controls.

[21] This assumes that β_2 generalizes beyond children with half-siblings. To support this, we show that cross-sectional parent-child associations are similar in both samples (Table A9) despite children with half-siblings being slightly negatively selected (Table A11). The estimates are also robust to reweighting the subsample so that mean parental income and age-at-first-birth matches the core sample (Table A12).

Extensions. Birth order may affect partner choice along unobserved dimensions not captured by partner income. If earlier-born children select more advantaged partners along these dimensions, our mediation analysis understates the role of partner choice. To address this, we aggregate all other information we have of partners into an index and repeat the two-step procedure with the index as an additional mediator.[22] The resulting estimates are similar to those obtained using partner income alone (Table A10). We conclude that although the index captures important dimensions of partner human capital not captured by income, changes in these characteristics cannot further explain the intergenerational effects.

Lastly, for comparison we also estimate the main specification including partner income and the extended family index as (bad) control variables (Table A10). The resulting estimates are very similar to those obtained with the two-step approach, but this should not be interpreted as evidence that naively controlling for partner characteristics is generally valid. Our specification includes family fixed effects, which prove sufficient to absorb much of the bias arising from assortative mating. In identification designs that rely on between-family variation, differences in partner characteristics may reflect stronger differences in unobserved traits of the treated parent and can therefore still result in substantial bias.[23]

6.2 Fertility

Another potentially important mediator is fertility. To examine this, we report birth order effects on the likelihood of having any children, the total number of children, and age at first birth in Table A8.

We find non-monotonic birth order effects on the likelihood of having any children.

[22] We use the average income and wealth ranks of partners' siblings and the wealth rank of their parents, which are among the strongest predictors of child income in the Netherlands (de Vries (2025)). We describe how we aggregate this into one index in Appendix B. This index better predicts child income than partner income and is only weakly correlated with partner income ($\rho = 0.13$).

[23] This pattern is also visible in Table 5. The cross-sectional estimates of β_2, which are based on between-family variation, are much larger than those based on parent fixed effects.

Table 5: Mediation Analysis for Partner Choice

	(1)	(2)	(3)
A. Birth order effects	Partner income (δ_1)	Child income (β)	Residualized child income (β_1)
Maternal birth order effect	-0.371***	-0.215***	-0.178***
	(0.036)	(0.055)	(0.066)
Paternal birth order effect	-0.474***	-0.319***	-0.269***
	(0.036)	(0.055)	(0.061)
	2,679,188	2,679,188	2,679,188
B. Mediator effects	Child income (cross-sectional)	Child income (fixed effects, β_2)	% effect due to partner choice ($1 - \beta_1/\beta$)
Income partner of the mother	0.262***	0.099***	17%
	(0.008)	(0.012)	
Income partner of the father	0.192***	0.091***	16%
	(0.008)	(0.011)	
Parent fixed effects		x	
N	49,569	49,569	

Notes: Panel A column 1 reports the effect of parental birth order on partner income rank, estimated according to equation 2, where birth order is interacted with gender. Column 2 presents results from the same specification using the child's income rank as the outcome. Column 3 reports the effect of parental birth order on residualized child income, estimated according to equation 8. The sample in panel A includes all children of the core sample for whom partner income data are available (93 percent). In column 3, standard errors are based on a clustered bootstrap with 200 replications. Panel B column 1 reports a regression of child income on the focal parent's gender, the income rank of the focal parent's partner, and their interaction. Column 2 adds fixed effects for the focal parent. The sample in panel B includes all children from panel A with half siblings. The focal parent is defined as the parent who has children with at least two partners. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Second-born individuals are slightly more likely to have children than firstborns, whereas for fourth- and fifth-borns this reverses. Such fertility differences could drive our results when the types of individuals who opt in or out of parenthood due to their birth order systematically differ. However, if selective fertility were a key driver, we would expect similarly non-monotonic patterns in our main outcomes, which we do not observe. Moreover, the effect sizes are very small: for instance, second-borns are only 0.8 percentage points more likely to have children than firstborns, and thirdborns just 0.3 percentage points more likely, relative to a mean of 78 percent.

Conditional on having children, we find monotone and sizable negative effects of birth order on the number of children and timing of first birth. As a result, children of higher birth-order parents are born earlier and have a lower birth order. These differences may mediate the main effect through cohort-specific trends or birth order effects in the next generation. As before, adding these mediators as control variables is problematic when differences in household size or birth timing also reflect differences in the parents' characteristics.

To address this, we present a second two-step estimator in Appendix C that solves for between-household confounding. The logic parallels our mediation analysis of partner choice. In the first step, we estimate how the mediators (birth order and year of birth) affect child income using only within-household variation. In the second step, we use these estimates to adjust child income for the indirect effects operating through these channels, and re-estimate a cousin fixed effects model with the residualized outcome.

Table 6 panel A replicates the main intergenerational birth order effect in column 1 and applies the two-step estimator in column 2. Using these estimates, column 3 shows that changes in the number and timing of children can explain 18 percent of the main effect.[24] This shows that, as for partner choice, these mediators explain an economically relevant share of the effects.

[24]We find similar estimates when we add children's birth year and birth order as (bad) controls (Table A13). Consistent with the results for partner choice, this suggests that the extended family fixed effects already absorb much of the between-household confounders.

Table 6: Fertility and Neighborhoods

A. Fertility	Child household income rank		% effect due to fertility
Birth order parent	-0.268***	-0.221**	18%
	(0.055)	(0.085)	
Two-step		x	
N	2,871,495	2,871,495	
B. Neighborhoods	Child household income rank		% effect due to neighborhoods
Birth order parent	-0.276***	-0.259***	6%
	(0.054)	(0.054)	
Neighborhood fixed effects		x	
N	2,807,379	2,807,379	

Notes: Panels A and B column 1 report the effect of parental birth order on child income, estimated according to equation 2. Column 2 of panel A uses the Two-Step estimator (see Appendix C), while column 2 of panel B adds neighborhood fixed effects to the main specification. Panel A uses the core analysis sample, and panel B drops children for whom neighborhoods are not observed (2.2 percent). Standard errors are in parentheses and clustered at the household level. In panel A, column 2, standard errors are based on a clustered bootstrap with 200 replications. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

6.3 Neighborhoods

The final mediating channel we consider is the neighborhood in which children grow up. A large literature shows that neighborhoods, even at a highly local level, have substantial effects on long-run income (Chyn and Katz (2021), Mogstad and Torsvik (2023)). If early-childhood advantages enables individuals to raise their children in high-opportunity neighborhoods, then neighborhoods may be an important channel through which advantage persists across generations. We are not aware of papers studying causal intergenerational effects that explicitly quantify this channel. We exploit very granular information on neighborhoods, with average and median sizes of 1,160 and 560 residents, to provide new insights.[25]

We first show in Table A14 that earlier-born parents sort into significantly higher-income neighborhoods. The point estimate is small (0.014 standard deviations), but neighborhood

[25]Neighborhood identifiers are first observed in 1995. We match children born in 1980 or later (69 percent) to their neighborhood in 1995. For earlier cohorts, we use the neighborhood in which the mother resides in 1995. This is a good proxy: For the 1980 cohort, for whom we observe both measures, the mother's neighborhood when the child is age 20 equals the child's neighborhood at age 16 in 82 percent of the cases.

sorting could still contribute to the observed intergenerational effect if parents also sort on unobserved dimensions of neighborhood quality.

We assess this by extending the Cousin Fixed Effects model with neighborhood fixed effects. A key advantage is that these fixed effects absorb all neighborhood characteristics, both observed and unobserved, affecting child income. A concern is that neighborhood choice may partly reflect parental characteristics, and we do not have additional exogenous variation to address this. We thus rely on the cousin fixed effects to address this bias. Reassuringly, in the mediation analyses for partner choice and fertility this proved sufficient.

Consistent with the small effects on neighborhood income, panel B of Table 6 shows that the intergenerational effect estimates fall by 6 percent when we add neighborhood fixed effects. We conclude that differences in neighborhood sorting explain only a minor share of the intergenerational effects.[26]

7 Discussion and Conclusion

In this paper, we provide insights into how parents' own childhood experiences affect the future of their children. Leveraging population data of two generations of Dutch families and the widespread applicability of birth order effects, we precisely estimate that parents transmit 18 percent of their income advantages due to their birth order to the next generation. We also find sizable effects on education and crime, underscoring that early-life advantages can have important persistent economic consequences.

Theory and evidence from Rhesus monkeys suggest that treatments in one generation can even improve *the returns* to parental investments into the next (Becker et al. (2018), Dettmer et al. (2026)). We exploit that birth order generates variation in parental investments across two generations to provide the first test for intergenerational complementarities among humans. We find no evidence that firstborn children disproportionally benefit from

[26]If higher ability parents sort into higher opportunity neighborhoods (as documented by Chetty et al. (Forthcoming)), the mediation analysis may even overstate the role of neighborhoods.

the improved human capital of their firstborn parents, suggesting that intergenerational complementarities are either limited, context-specific, or offset by changes in the level of parental investments.

Lastly, we use novel methods to show that changes in partner choice (17 percent), fertility (18 percent), and neighborhoods (6 percent) jointly explain an important share of the transmitted advantages. Quantifying their mediating role serves two purposes. First, it helps understand through which channels effects persist across generations. Second, it informs whether intergenerational causal effects represent net social gains. Our results suggest that an economically relevant share of the total effect does not necessarily reflect such gains, since improvements in partner or neighborhood quality for treated individuals can come at the expense of others. If similar mechanisms underlie the intergenerational effects of policy interventions such as extended education or preschool attendance, naive intergenerational effect estimates may overstate their true social returns.

Going beyond birth order, we view the three ingredients above as crucial for evaluating the intergenerational benefits of any treatment, especially policy interventions: (i) a precise estimate of the magnitude of intergenerational benefits, (ii) an assessment of whether these benefits reflect net societal gains, and (iii) an examination of whether combining treatments amplifies their returns. Accumulating evidence on these three components across multiple childhood experiences can help policymakers design interventions to effectively 'break the cycle' of income persistence.

References

Abrahamsson, Sara Sofie, Aline Bütikofer, Katrine V. Løken, and Marianne E. Page. 2025. "Sources of Generational Persistence in the Effects of Early-Life Health Interventions." National Bureau of Economic Research Working Paper 33612.

Akgündüz, Yusuf, Pelin Akyol, Abdurrahman Aydemir, Murat Demirci, and Murat G. Kïrdar. 2024. "Intergenerational Effects of Compulsory Schooling Reform on Early Childhood Development in a Middle-Income Country." IZA Discussion Paper 17249.

Akresh, Richard, Sonia Bhalotra, Marinella Leone, and Una Osili. 2023a. "First- and Second-Generation Impacts of the Biafran War." *Journal of Human Resources* 58 (2): 488–531.

Akresh, Richard, Daniel Halim, and Marieke Kleemans. 2023b. "Long-Term and Intergenerational Effects of Education: Evidence from School Construction in Indonesia." *The Economic Journal* 133 (650): 582–612.

Alesina, Alberto, Stefanie Stantcheva, and Edoardo Teso. 2018. "Intergenerational Mobility and Preferences for Redistribution." *American Economic Review* 108 (2): 521–554.

Almond, Douglas, Janet Currie, and Valentina Duque. 2018. "Childhood Circumstances and Adult Outcomes: Act II." *Journal of Economic Literature* 56 (4): 1360–1446.

Angrist, Joshua, Victor Lavy, and Analia Schlosser. 2010. "Multiple Experiments for the Causal Link between the Quantity and Quality of Children." *Journal of Labor Economics* 28 (4): 773–824.

Averett, Susan L., Laura M. Argys, and Daniel I. Rees. 2011. "Older Siblings and Adolescent Risky Behavior: Does Parenting Play a Role?" *Journal of Population Economics* 24 (3): 957–978.

Barclay, Kieron, Torkild Lyngstad, and Dalton Conley. 2021. "The Production of Inequalities within Families and across Generations: The Intergenerational Effects of Birth Order on Educational Attainment." *European Sociological Review* 37 (4): 607–625.

Barr, Andrew, and Chloe R. Gibbs. 2022. "Breaking the Cycle? Intergenerational Effects of an Antipoverty Program in Early Childhood." *Journal of Political Economy* 130 (12): 3253–3285.

Barrios Fernández, Andrés, Christopher Neilson, and Seth Zimmerman. 2024. "Elite Universities and the Intergenerational Transmission of Human and Social Capital." IZA Discussion Paper 17252.

Becker, Gary S., Scott Duke Kominers, Kevin M. Murphy, and Jörg L. Spenkuch. 2018. "A Theory of Intergenerational Mobility." *Journal of Political Economy* 126 (S1): S7–S25.

Bharadwaj, Prashant, N. Meltem Daysal, and Miriam Wüst. 2022. "Spillover Effects of Early-Life Medical Interventions." *The Review of Economics and Statistics* 104 (4): 796–811.

Black, Nicole, Danusha Jayawardana, and Gawain Heckley. 2026. "The effect of birth order on children's time use." *Journal of Economic Behavior & Organization* 242 107418.

Black, Sandra E., Aline Bütikofer, Paul J. Devereux, and Kjell G. Salvanes. 2019. "This Is Only a Test? Long-Run and Intergenerational Impacts of Prenatal Exposure to

Radioactive Fallout." *The Review of Economics and Statistics* 101 (3): 531–546.

Black, Sandra E., Paul J. Devereux, and Kjell G. Salvanes. 2005a. "Why the Apple Doesn't Fall Far: Understanding Intergenerational Transmission of Human Capital." *American Economic Review* 95 (1): 437–449.

Black, Sandra E., Paul J. Devereux, and Kjell G. Salvanes. 2005b. "The More the Merrier? The Effect of Family Size and Birth Order on Children's Education." *The Quarterly Journal of Economics* 120 (2): 669–700.

Black, Sandra E., Paul J. Devereux, and Kjell G. Salvanes. 2011. "Older and Wiser? Birth Order and IQ of Young Men." *CESifo Economic Studies* 57 (1): 103–120.

Black, Sandra E., Erik Grönqvist, and Björn Öckert. 2018. "Born to Lead? The Effect of Birth Order on Noncognitive Abilities." *The Review of Economics and Statistics* 100 (2): 274–286.

Breining, Sanni, Joseph Doyle, David N. Figlio, Krzysztof Karbownik, and Jeffrey Roth. 2020. "Birth Order and Delinquency: Evidence from Denmark and Florida." *Journal of Labor Economics* 38 (1): 95–142.

Brenøe, Anne Ardila, and Ramona Molitor. 2018. "Birth Order and Health of Newborns." *Journal of Population Economics* 31 (2): 363–395.

Bütikofer, Aline, and Kjell G Salvanes. 2020. "Disease Control and Inequality Reduction: Evidence from a Tuberculosis Testing and Vaccination Campaign." *The Review of Economic Studies* 87 (5): 2087–2125.

Carneiro, Pedro, Costas Meghir, and Matthias Parey. 2013. "Maternal Education, Home Environments, and the Development of Children and Adolescents." *Journal of the European Economic Association* 11 123–160.

Chadwick, Laura, and Gary Solon. 2002. "Intergenerational Income Mobility Among Daughters." *American Economic Review* 92 (1): 335–344.

Chetty, Raj, John N Friedman, Nathaniel Hendren, Maggie R Jones, and Sonya R Porter. Forthcoming. "The Opportunity Atlas: Mapping the Childhood Roots of Social Mobility." *American Economic Review*.

Chetty, Raj, John N Friedman, Nathaniel Hilger, Emmanuel Saez, Diane Whitmore Schanzenbach, and Danny Yagan. 2011. "How does your kindergarten classroom affect your earnings? Evidence from Project STAR." *The Quarterly journal of economics* 126 (4): 1593–1660.

Chetty, Raj, John N Friedman, and Jonah E Rockoff. 2014. "Measuring the impacts of teachers II: Teacher value-added and student outcomes in adulthood." *American economic review* 104 (9): 2633–2679.

Chetty, Raj, Nathaniel Hendren, and Lawrence F Katz. 2016. "The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment." *American Economic Review* 106 (4): 855–902.

Chevalier, Arnaud, Colm Harmon, Vincent O' Sullivan, and Ian Walker. 2013. "The Impact of Parental Income and Education on the Schooling of Their Children." *IZA Journal of Labor Economics* 2 (1): 8.

Chyn, Eric, and Lawrence F. Katz. 2021. "Neighborhoods Matter: Assessing the Evidence for Place Effects." *Journal of Economic Perspectives* 35 (4): 197–222.

Collado, M Dolores, Ignacio Ortuño-Ortín, and Jan Stuhler. 2023. "Estimating Intergenerational and Assortative Processes in Extended Family Data." *The Review of*

Economic Studies 90 (3): 1195–1227.

Cunha, Flavio, and James Heckman. 2007. "The Technology of Skill Formation." *American Economic Review* 97 (2): 31–47.

Currie, Janet, and Enrico Moretti. 2003. "Mother's Education and the Intergenerational Transmission of Human Capital: Evidence from College Openings." *The Quarterly Journal of Economics* 118 (4): 1495–1532.

Dahl, Gordon B., and Anne Gielen. 2024. "Persistent Effects of Social Program Participation on the Third Generation." IZA Discussion Paper 16816.

Dahl, Gordon B., Katrine V. Løken, and Magne Mogstad. 2014. "Peer Effects in Program Participation." *American Economic Review* 104 (7): 2049–2074.

De Haan, Monique. 2011. "The Effect of Parents' Schooling on Child's Schooling: A Nonparametric Bounds Analysis." *Journal of Labor Economics* 29 (4): 859–892.

Dettmer, Amanda M, James J Heckman, Juan Pantano, Victor Ronda, and Stephen J Suomi. 2026. "Effects of Multigenerational Exposure to Early-Life Advantage: Lessons from a Primate Study." *Journal of Political Economy* 134 (1): 285–312.

Domingue, Benjamin W., Daniel Belsky, Dalton Conley, Kathleen Mullan Harris, and Jason D. Boardman. 2015. "Polygenic Influence on Educational Attainment: New Evidence from The National Longitudinal Study of Adolescent to Adult Health." *AERA open* 1 (3): 1–13.

East, Chloe N., Sarah Miller, Marianne Page, and Laura R. Wherry. 2023. "Multigenerational Impacts of Childhood Access to the Safety Net: Early Life Exposure to Medicaid and the Next Generation's Health." *American Economic Review* 113 (1): 98–135.

Ermisch, John, Marco Francesconi, and Thomas Siedler. 2006. "Intergenerational Mobility and Marital Sorting." *The Economic Journal* 116 (513): 659–679.

Fernández, Raquel, and Richard Rogerson. 2001. "Sorting and Long-Run Inequality." *The Quarterly Journal of Economics* 116 (4): 1305–1341.

García, Jorge Luis, James J. Heckman, and Victor Ronda. 2023. "The Lasting Effects of Early-Childhood Education on Promoting the Skills and Social Mobility of Disadvantaged African Americans and Their Children." *Journal of Political Economy* 131 (6): 1477–1506.

Haider, Steven, and Gary Solon. 2006. "Life-Cycle Variation in the Association between Current and Lifetime Earnings." *American Economic Review* 96 (4): 1308–1320.

Havari, Enkelejda, and Marco Savegnago. 2022. "The Intergenerational Effects of Birth Order on Education." *Journal of Population Economics* 35 (1): 349–377.

Holmlund, Helena. 2022. "How Much Does Marital Sorting Contribute to Intergenerational Socioeconomic Persistence?" *Journal of Human Resources* 57 (2): 372–399.

Holmlund, Helena, Mikael Lindahl, and Erik Plug. 2011. "The Causal Effect of Parents' Schooling on Children's Schooling: A Comparison of Estimation Methods." *Journal of Economic Literature* 49 (3): 615–651.

Hotz, V. Joseph, and Juan Pantano. 2015. "Strategic Parenting, Birth Order, and School Performance." *Journal of Population Economics* 28 (4): 911–936.

Houmark, Mikkel Aagaard. 2023. "First Among Equals? How Birth Order Shapes Child Development." MPRA Paper 119325.

Ilciukas, Julius, Petter Lundborg, Erik Plug, and Astrid Würtz Rasmussen. 2025. "The Only Child." IZA Discussion Paper No. 17641.

Inoue, Atsushi, and Gary Solon. 2010. "Two-Sample Instrumental Variables Estimators." *The Review of Economics and Statistics* 92 (3): 557–561.

Isungset, Martin Arstad, Jeremy Freese, Ole A Andreassen, and Torkild Hovde Lyngstad. 2022. "Birth Order Differences in Education Originate in Postnatal Environments." *PNAS Nexus* 1 (2): pgac051.

Kantarevic, Jasmin, and Stéphane Mechoulan. 2006. "Birth Order, Educational Attainment, and Earnings An Investigation Using the PSID." *Journal of Human Resources* 41 (4): 755–777.

Lehmann, Jee-Yeon K., Ana Nuevo-Chiquero, and Marian Vidal-Fernandez. 2018. "The Early Origins of Birth Order Differences in Children's Outcomes and Parental Behavior." *Journal of Human Resources* 53 (1): 123–156.

Lin, Wanchuan, Juan Pantano, and Shuqiao Sun. 2020. "Birth order and unwanted fertility." *Journal of Population Economics* 33 (2): 413–440.

Lundborg, Petter, Anton Nilsson, and Dan-Olof Rooth. 2014. "Parental Education and Offspring Outcomes: Evidence from the Swedish Compulsory School Reform." *American Economic Journal: Applied Economics* 6 (1): 253–278.

Maurin, Eric, and Sandra McNally. 2008. "Vive La Révolution! Long-Term Educational Returns of 1968 to the Angry Students." *Journal of Labor Economics* 26 (1): 1–33.

Mazumder, Bhashkar. 2005. "Fortunate Sons: New Estimates of Intergenerational Mobility in the United States Using Social Security Earnings Data." *The Review of Economics and Statistics* 87 (2): 235–255.

Mazumder, Bhashkar, Maria Fernanda Rosales-Rueda, and Margaret Triyana. 2023. "Social Interventions, Health, and Well-Being: The Long-Term and Intergenerational Effects of a School Construction Program." *Journal of Human Resources* 58 (4): 1097–1140.

McCrary, Justin, and Heather Royer. 2011. "The Effect of Female Education on Fertility and Infant Health: Evidence from School Entry Policies Using Exact Date of Birth." *American Economic Review* 101 (1): 158–195.

Mogstad, Magne, and Gaute Torsvik. 2023. "Family Background, Neighborhoods, and Intergenerational Mobility." In *Handbook of the Economics of the Family*, edited by Lundberg, Shelly, and Alessandra Voena Volume 1. 327–387.

Muslimova, Dilnoza, Hans van Kippersluis, Cornelius A. Rietveld, Stephanie von Hinke, and Fleur Meddens. Forthcoming. "Gene-Environment Complementarity in Educational Attainment." *Journal of Labor Economics*.

Nicoletti, Cheti, Kjell G. Salvanes, and Emma Tominey. 2018. "The Family Peer Effect on Mothers' Labor Supply." *American Economic Journal: Applied Economics* 10 (3): 206–234.

Nybom, Martin, and Jan Stuhler. 2017. "Biases in Standard Measures of Intergenerational Income Dependence." *The Journal of Human Resources* 52 (3): 800–825.

Oreopoulos, Philip, Marianne E. Page, and Ann Huff Stevens. 2006. "The Intergenerational Effects of Compulsory Schooling." *Journal of Labor Economics* 24 (4): 729–760.

Page, Marianne E. Forthcoming. "New Advances on an Old Question: Does Money Matter for Children's Outcomes?" *Journal of Economic Literature*.

Pavan, Ronni. 2016. "On the Production of Skills and the Birth-Order Effect." *Journal of Human Resources* 51 (3): 699–726.

Pekkarinen, Tuomas, Roope Uusitalo, and Sari Kerr. 2009. "School Tracking and Intergenerational Income Mobility: Evidence from the Finnish Comprehensive School Reform." *Journal of Public Economics* 93 (7): 965–973.

Price, Joseph. 2008. "Parent-Child Quality Time: Does Birth Order Matter?" *Journal of Human Resources* 43 (1): 240–265.

Rossin-Slater, Maya, and Miriam Wüst. 2020. "What Is the Added Value of Preschool for Poor Children? Long-Term and Intergenerational Impacts and Interactions with an Infant Health Intervention." *American Economic Journal: Applied Economics* 12 (3): 255–286.

Sikhova, Aiday. 2023. "Understanding the Effect of Parental Education and Financial Resources on the Intergenerational Transmission of Income." *Journal of Labor Economics* 41 (3): 771–811.

de Vries, Sander. 2025. "Who Gets Ahead? Measuring Income Gaps across Family Backgrounds." Tinbergen Institute Discussion Paper 2025-010/V.

Walker, Michael W., Alice H. Huang, Suleiman Asman et al. 2023. "Intergenerational Child Mortality Impacts of Deworming: Experimental Evidence from Two Decades of the Kenya Life Panel Survey." National Bureau of Economic Research Working Paper 31162.

Appendix A: Supplementary Results

Table A1: Birth order effects for all individuals and parents in our sample only

Birth Order	Income	
	(1)	(2)
2	-1.577***	-1.502***
	(0.053)	(0.097)
3	-2.958***	-2.799***
	(0.094)	(0.169)
4	-4.106***	-3.708***
	(0.155)	(0.274)
5	-5.162***	-4.853***
	(0.258)	(0.445)
Mean	53.766	53.599
SD	27.901	27.394
Sample	All individuals	Parents sample
N	3,241,987	1,465,530

Notes: This table presents birth order effects for the first generation for two samples (equation 1). Column 1 estimates birth order effects for the entire first generation sample, replicating the main result in Table 2. Column 2 applies the same regression to the subsample of individuals with children born before 1991. These are all the parents of the children in our core analysis sample for the intergenerational effects. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A2: (Parental) birth order effects with alternative income measures

(Parental)	Household Income Rank		Log Household Income		Personal Income Rank	
Birth Order	(1)	(2)	(3)	(4)	(5)	(6)
2	-1.590***	-0.307***	-0.033***	-0.007***	-1.144***	-0.257***
	(0.053)	(0.066)	(0.001)	(0.002)	(0.045)	(0.065)
3	-2.971***	-0.538***	-0.063***	-0.012***	-2.263***	-0.515***
	(0.094)	(0.115)	(0.002)	(0.003)	(0.080)	(0.114)
4	-4.125***	-0.708***	-0.089***	-0.016***	-3.123***	-0.710***
	(0.154)	(0.187)	(0.004)	(0.005)	(0.132)	(0.185)
5	-5.219***	-0.820***	-0.118***	-0.018**	-4.019***	-0.781***
	(0.258)	(0.304)	(0.006)	(0.007)	(0.219)	(0.301)
Mean	53.91	55.016	11.319	11.306	53.076	54.537
SD	27.799	27.171	0.67	0.655	28.504	27.282
Generation	1	2	1	2	1	2
N	3,233,273	2,868,930	3,233,273	2,868,930	3,233,273	2,868,930

Notes: This table presents birth order effects (equation 1) and intergenerational birth order effects (equation 2) on various income measures. The samples include all children (99.9%) and parents (99.7%) whose personal incomes are observed. Columns 1 and 2 replicate the main result for these subsamples. Columns 3 and 4 report results using the log of household income. Columns 5 and 6 report results using the personal income rank, which is computed relative to all individuals in the same cohort and of the same gender. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A3: (Parental) birth order effects with varying income observations

	A. Birth order effects				
Birth order	-1.467***	-1.485***	-1.478***	-1.467***	-1.462***
	(0.044)	(0.044)	(0.044)	(0.044)	(0.044)
# income observations	1	2	3	4	5
N	3,241,987	3,241,987	3,241,987	3,241,987	3,241,987
	B. Intergenerational birth order effects				
Parental birth order	-0.248***	-0.252***	-0.267***	-0.269***	-0.265***
	(0.055)	(0.055)	(0.055)	(0.055)	(0.055)
# income observations	1	2	3	4	5
N	2,848,041	2,848,041	2,848,041	2,848,041	2,848,041
	C. Intergenerational birth order effects				
Parental birth order	-0.269***	-0.290***			
	(0.055)	(0.055)			
Only income above age 32	No	Yes			
N	2,848,041	2,848,041			

Notes: Panel A (B) estimates (intergenerational) birth order effects on lifetime income according to equation 1 (2), where lifetime income is proxied using a varying number of income observations. Panel C column (1) estimates intergenerational birth order effects using the same lifetime income variable as used in the main analysis, whereas column (2) replicates it using only incomes above age 32. This restriction reduces the number of income observations for cohorts born between 1988 and 1990 to fewer than four. Standard errors in panels B and C are clustered on the household level. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A4: Parental birth order effects on education: estimated by family size

Parental Birth Order	All	Family size 2	Family size 3	Family size 4	Family size 5
	(1)	(2)	(3)	(4)	(5)
2	-0.750***	-0.981***	-0.442**	-0.877***	-1.290***
	(0.113)	(0.268)	(0.195)	(0.209)	(0.264)
3	-1.502***		-1.769***	-1.017***	-1.834***
	(0.197)		(0.368)	(0.324)	(0.361)
4	-2.541***			-1.961***	-3.043***
	(0.320)			(0.511)	(0.512)
5	-2.488***				-3.129***
	(0.519)				(0.731)
Mean	39.953	39.711	40.302	40.139	39.458
SD	48.98	48.93	49.05	49.018	48.876
N	2,952,774	878,239	906,534	698,895	469,106

Notes: This table presents the effect of parental birth order on children's higher education attainment. The estimates are separated by family size. The sample includes all children from the core analysis sample, including those with missing income. All models are estimated according to equation 2. Standard errors are clustered on the household level and reported in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A5: Parental birth order effects on boys' criminal behavior: estimated by family size

Parental Birth Order	All	Family size 2	Family size 3	Family size 4	Family size 5
	(1)	(2)	(3)	(4)	(5)
2	0.327**	0.758**	0.160	0.375	0.282
	(0.135)	(0.350)	(0.213)	(0.248)	(0.358)
3	0.708***		0.662	0.555	0.550
	(0.241)		(0.418)	(0.378)	(0.455)
4	1.199***			0.869	1.257**
	(0.395)			(0.600)	(0.637)
5	1.451**				1.460
	(0.649)				(0.908)
Mean	9.549	9.481	9.374	9.755	9.927
SD	29.389	29.296	29.146	29.67	29.902
N	1,521,986	549,259	501,758	305,275	165,694

Notes: This table presents the effect of parental birth order on boys' criminal behavior. The outcome is an indicator which equals one if a child is suspected of any crime between ages 18 to 20. The estimates are separated by family size. All models are estimated according to equation 2. Standard errors are clustered on the household level and reported in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A6: Parental birth order effects on different types of crime

Birth Order Parent	Crime	Violent Crime	Property Crime	Other Crime
2	0.327**	0.190**	0.142*	0.058
	(0.135)	(0.090)	(0.081)	(0.101)
3	0.708***	0.445***	0.195	0.232
	(0.241)	(0.161)	(0.143)	(0.178)
4	1.199***	0.609**	0.343	0.428
	(0.395)	(0.266)	(0.237)	(0.292)
5	1.451**	0.761*	0.583	0.141
	(0.649)	(0.440)	(0.393)	(0.479)
Mean	9.549	3.998	3.121	4.943
SD	29.389	19.592	17.388	21.677
N	1,521,986	1,521,986	1,521,986	1,521,986

Notes: This table presents the effects of parental birth order on children's likelihood to be suspected of any crime between ages 18 to 20, as well as for three (non-mutually exclusive) categories: property crime, violent crime, and 'other' crimes that do not fit those categories. All models are estimated according to equation 2. Standard errors are clustered on the household level and reported in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A7: Intergenerational Complementarities by Gender

	Income	Higher education	Crime
Birth order	-1.226***	-3.899***	1.199***
	(0.146)	(0.256)	(0.366)
Birth order parent	-0.244***	-0.770***	0.201*
	(0.064)	(0.109)	(0.124)
Birth order × birth order parent	0.045	-0.023	0.162
	(0.061)	(0.113)	(0.169)
Birth order × father	-0.160	-0.561	0.272
	(0.218)	(0.383)	(0.506)
Birth order parent × father	-0.068	-0.127	0.230**
	(0.06)	(0.107)	(0.103)
Birth order × birth order parent ×	0.009	0.175	-0.212
father	(0.094)	(0.164)	(0.224)
N	2,871,495	2,871,495	1,521,986

Notes: each column reports birth order effects, parental birth order effects, and their interaction effect, estimated according to equation 3 with all variables interacted by gender of the parent. Column 1 presents results for household income ranks. Column 2 presents results for higher education completion. The outcome in column 3 is an indicator for whether a son is suspected of a crime between ages 18 and 20. The sample in Column 3 includes only sons. Standard errors are in parentheses. The standard errors are based on a clustered bootstrap with 200 replications. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A8: Birth order effects on fertility

Birth Order	Has child	Number of children	Age at first child
	(1)	(2)	(3)
2	0.829***	-0.019***	-0.132***
	(0.084)	(0.002)	(0.012)
3	0.353**	-0.044***	-0.253***
	(0.149)	(0.004)	(0.021)
4	-0.521**	-0.068***	-0.342***
	(0.247)	(0.006)	(0.034)
5	-1.966***	-0.098***	-0.510***
	(0.413)	(0.011)	(0.057)
Mean	78.494	2.198	28.823
SD	41.086	0.868	5.278
N	3,241,987	2,544,771	2,544,771

Notes: This table presents the effect of birth order on fertility outcomes, estimated according to equation 1. The outcome in column 1 is an indicator for whether an individual has at least one child, and the sample includes the core analysis sample of the first generation. We have converted the estimates to percentage points for readability. The outcomes in columns 2 and 3 measure the number of children and the age when parents have their first child. The sample in columns 2 and 3 includes all individuals from the first generation who have a child. Standard errors are in parentheses (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A9: Effect of partner income and the partner family index on child income

	Household income rank		
	(1)	(2)	(3)
Income partner of the mother	0.252***	0.226***	0.092***
	(0.001)	(0.008)	(0.012)
Income partner of the father	0.092***	0.142***	0.081***
	(0.001)	(0.008)	(0.011)
Family index partner of the mother	0.705***	0.745***	0.306***
	(0.005)	(0.041)	(0.060)
Family index partner of the father	0.969***	1.063***	0.494***
	(0.005)	(0.034)	(0.050)
Parent fixed effects			x
Sample	Core sample	Half-siblings	Half-siblings
N	2,679,188	49,569	49,569

Notes: Each column reports a regression of child income on partner income and the partner family index, both of which are interacted by gender of the treated parent. Columns 2 and 3 add parent fixed effects. The sample in column 1 includes all children from the core sample with non-missing partner income (93 percent). Columns 2 and 3 include only the subset of children with a half-sibling. The parent fixed effects correspond to the parent who has children with at least two partners. Standard errors are in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A10: Intergenerational birth order effects while controlling for partner income and extended family

	Child household income rank				
	(1)	(2)	(3)	(4)	(5)
Maternal birth order effects	-0.215***	-0.178***	-0.178***	-0.177***	-0.177***
	(0.055)	(0.055)	(0.066)	(0.055)	(0.066)
Paternal birth order effects	-0.319***	-0.253***	-0.269***	-0.239***	-0.255***
	(0.055)	(0.055)	(0.061)	(0.055)	(0.061)
Control for partner income		x			
Two-step for partner income			x		
Control for partner family index				x	
Two-step for partner income and family index					x
	2,679,188	2,679,188	2,679,188	2,679,188	2,679,188

Notes: This table presents parental birth order effects while controlling for the income rank and/or partner family index of the non-focal parent. Model 1 replicates the main result using equation 2 interacted by parent gender. Column 2 includes the partner's income rank as a control variable. Column 3 uses the two-step approach (equation 8). Column 4 includes partner income and the partner family index as controls. Column 5 uses the two-step approach with the partner family index as additional mediator. The sample includes all children from the core sample for whom the non-treated parent's personal income rank is available. The standard errors in parentheses are clustered at the household level. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A11: Descriptive statistics of children with half-siblings and their parents

	Fathers			Mothers		
	(1)	(2)	(3)	(1)	(2)	(3)
Age at first child	27.1	23.7	27.1	24.9	21.3	25.0
Personal income rank parent	71.8	60.8	71.9	29.8	29.9	29.9
Household income rank child	55.6	44.9	49.6	55.5	43.8	48.2
N	1,232,427	27,801	26,766	1,397,192	21,768	21,318
Sample	Full sample	Half siblings	Half siblings	Full sample	Half siblings	Half siblings
Weighted			x			x

Notes: This table reports descriptive statistics for all children in the full sample and for those with half-siblings. The table is split by the gender of the focal parent, defined as the parent from the first generation whose birth order effects are studied. Column 1 presents means for the full sample, column 2 for the subsample of children with half-siblings, and column 3 for the same subsample with reweighting. The weights in column 3 are constructed as follows. First, 100,000 children without half-siblings are randomly drawn from the full sample and combined with the half-sibling sample. Second, using this combined sample, the probability $\hat{p}$ that a child has a half-sibling is estimated from a logit model that includes the focal parent's age at first birth and income rank. Third, the weight is defined as $w = (1 - \hat{p})/\hat{p}$. This procedure upweights children whose focal parent has characteristics similar to those of parents in the full sample. Column 3 indicates that the weighting is successful, as the weighted means of parental age and income are equal to those in column 1.

Table A12: The effect of partner income on child income

	Child household income rank		
Income partner of the mother	0.099***	0.094***	0.125***
	(0.012)	(0.012)	(0.011)
Income partner of the father	0.091***	0.092***	0.094***
	(0.011)	(0.011)	(0.012)
Parent fixed effects	x	x	x
Controls		x	
Weigh observations			x
N	49,569	49,569	48,084

Notes: This table reports regressions of child income on partner income rank, interacted with the gender of the focal parent. All regressions include parent fixed effects. The sample consists of all children from the core sample who have a half-sibling in the core-analysis sample. Column 1 replicates the specification from Table 6 panel B column 2. Column 2 adds controls for the partner's year of birth, the child's year of birth, and relationship order (whether the child was born in the focal parent's first or a later relationship). Column 3 reports a weighted regression, with weights defined as in Table A11. Standard errors are in parentheses.(*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A13: Controlling for children's birth order and year of birth

	Child household income rank	
	(1)	(2)
Birth order parent	-0.268***	-0.211***
	(0.055)	(0.055)
Control for child birth order and year of birth		x
N	2,871,495	2,871,495

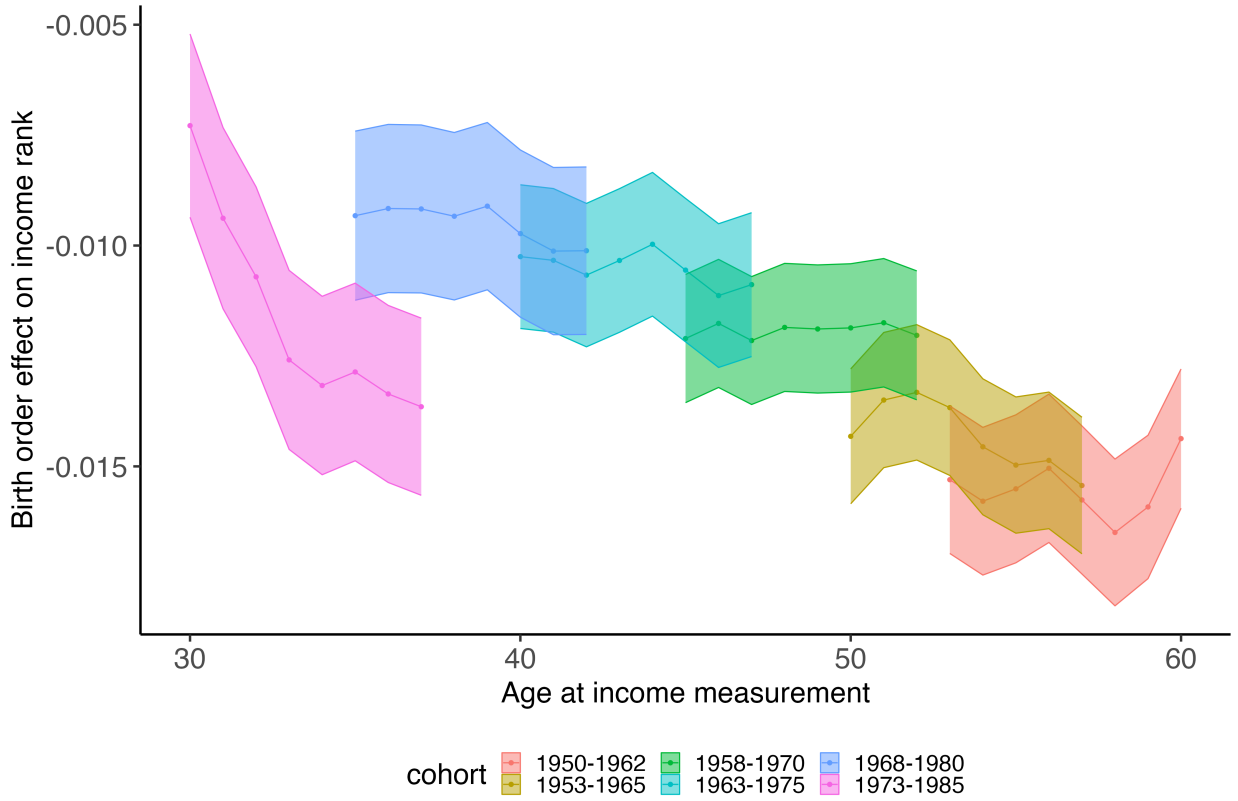
Notes: Column 1 replicates the main result using equation 2. Column 2 includes the children's year of birth and birth order as control variables. Standard errors are in parentheses. The standard errors are clustered at the household level. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Table A14: The effect of parental birth order on neighborhood income

	Neighborhood income
Parental birth order	-0.106***
	(0.009)
Mean	52.727
SD	7.359
N	2,807,379

Notes: This table presents the effect of parental birth order on the average neighborhood income rank, estimated according to equation 1. Neighborhoods are based on Statistics Netherlands' most granular neighborhood classifier (in Dutch: 'buurt'), with average and median neighborhood sizes of 1160 and 560 residents, respectively. For children born after 1979 (69%), we use the child's neighborhood in 1995. For the remaining children we use the mother's neighborhood in 1995. Average neighborhood income is measured by taking the average lifetime income ranks of all residents excluding the parents in 1995. The sample includes all children from the core analysis sample whose neighborhoods are observed (98 percent). Standard errors are clustered on the household level and reported in parentheses. (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$)

Figure A1: Birth order effects on income measured at different ages



Notes: This figure displays the estimated effect of birth order on income across cohorts and at different ages of income measurement. The Y-axis shows coefficients from Sibling Fixed Effects models (equation 1) where birth order is included as a numerical variable. The X-axis indicates the age at which income is observed. Each line corresponds to a different birth cohort, as denoted by color. Within each cohort, estimates are based on identical samples. Shaded bands represent 95% confidence intervals.

Appendix B: Constructing an Extended Family Index

This section explains how we construct the extended family index used in section 6.

We first link partners to their parents using the parent-child register. This is possible for 86 percent of partners. These parent identifiers allow us to link partners to their siblings.

We merge this information with household income ranks of the siblings. We use the same household income rank definition as in Section 2. For each partner, we compute the average household income rank across all observed siblings. Overall, we observe sibling income information for 80 percent of partners.

We next construct measures based on household wealth. Wealth data come from the wealth register and record the value of each asset and liability held by the household in 2006. We define net wealth as total assets minus total liabilities. Individuals are ranked by net wealth relative to others in the same cohort. As for income, we compute the average net wealth rank across all observed siblings. This yields sibling wealth information for 80 percent of partners.

Finally, we link net wealth ranks to the father and the mother of the partner when available. If both parents are observed, we take the average of their ranks. If only one parent is observed, we use that rank. This procedure yields a parental wealth rank for 60 percent of partners.

We combine the available information into an index as follows. For each of the three variables, we replace missing values with -1. We also create three indicators equal to one if the corresponding value is missing and zero otherwise. This approach preserves observations with missing values and uses missingness as additional information.[27] We then estimate a regression of child income rank on the three variables and the corresponding indicators using the core analysis sample, excluding all children with half-siblings.

The index equals the predicted income from this regression model. A higher value indicates more favorable extended family characteristics associated with child income. The index strongly predicts child income. In the core analysis sample, its explanatory power is $R^2 = 0.034$, which exceeds the $R^2 = 0.014$ obtained when using partner income alone. The correlation between the two measures is 0.13. This indicates that the index captures information about partner human capital not reflected in partner income alone. Lastly, we generate predictions for children with half-siblings, with whom we estimate the parent fixed effects models.

[27] For 85% of partners we have information of at least one outcome. The remaining 15% are also included in the analysis, but contribute little to the explanatory power of the index.

Appendix C: Decomposing the Fertility Effects

In Table A8 we show that children of parents with a higher birth order tend to be born later and have a lower birth order. In the presence of birth order effects or differences in time trends, these differences in birth order and year of birth are mediating factors that affect the outcome. Ideally, we would like to compare children of similar birth order and birth year. However, directly controlling for children's year of birth or birth order leads to a bad control problem because these are 'after-treatment' variables. To deal with this, we propose an estimator that allows us to estimate the intergenerational effect of birth order net of a child's own year of birth or birth order.

This two-step estimator is useful for two reasons. First, it can be used to determine whether time trends significantly affect the results. Although in our application the differences when using the two-step estimator are relatively small, they could be particularly important in situations where researchers find small intention-to-treat (ITT) effects and low take-up of the treatment. By inflating the ITT estimates by the take-up, any small differences in the year of birth will also be inflated, leading to potentially large differences in the total treatment effect.[28] Second, by normalizing all outcomes to the same birth order, the estimator allows researchers to use children of all birth orders, even in cases where treatment affects the number of children that parents have or when some children are censored. Using children of all birth orders maximizes the power and external validity of the estimates.

The formal set-up. A child c from parent p has birth order $BO^2_{cp} \in \{1, ..., B\}$, is born in year $t_{cp} \in \{1, ..., T\}$ and has income Y_{cp}. Let BO^1_p denote birth order of the treated parent p. As in Section 6, we omit all controls (cousin and cohort fixed effects) for notational convenience. All variables and error terms should be interpreted as residualized with respect to the full set of controls.

By assumption, the regression of Y_{cp} on BO^1_p estimates the total treatment effect β, which includes the mediating effects of birth order and year of birth. To decompose the effects into a part which can be explained by these mediators and a part which cannot, we consider the following causal model:

$$\begin{aligned} Y_{cp} &= \beta_1 BO^1_p + \underbrace{\sum_{b=1}^{B} \gamma_k I[BO^2_{cp} = b]}_{\gamma_{cp}} + \underbrace{\sum_{t=1}^{T} \tau_t I[t_{cp} = t]}_{\tau_{cp}} + \nu_p + \epsilon_{cp} \\ &= \beta_1 BO^1_p + \gamma_{cp} + \tau_{cp} + \nu_p + \epsilon_{cp}, \end{aligned} \tag{9}$$

where γ_{cp} and τ_{cp} are birth-order and year-of-birth fixed effects. By including dummies for each birth order and year of birth, the model above allows for non-linearity in their effects.

[28] For example, Rossin-Slater and Wüst (2020) find that women with access to preschool are 0.11 years older at their first birth. They also find that a mother's access to preschool at age 3 increases the likelihood that her child obtains more than a compulsory education by 0.9 percentage points. When inflated by the average take-up of 10 percent, their average treatment effect corresponds to roughly 10 percentage points. The 0.11 years difference in year of birth is also inflated by a factor of ten, which implies that the exposed children are born more than a year later on average. In the presence of strong positive trends in education, this could potentially explain a sizeable fraction of the total treatment effect.

The term β_1 captures the effect of parental birth order net of a child's year of birth and birth order. The error term ν_p represents unobserved characteristics that vary between families and are unrelated to parental birth order, and e_{cp} captures child-specific shocks.

Adding child birth order and birth year as controls is problematic because differences in the number or timing of children can be correlated with unobserved characteristics of parents that differ between families. For example, parents with higher education may on average have children later, so a child's year-of-birth (t_{cp}) would be correlated with differences in education between families (ν_p).

To address this issue, we exploit within-family variation in the second generation to estimate birth order and birth year effects. Because these effects are identified within families, the estimates are not confounded by unobserved characteristics that vary across families, such as parental income or education. As in Section 3, we assume that birth order and birth year effects are consistently estimated within families. Under this assumption, the two-step procedure proceeds as follows:

1. First, note that

$$Y_{cp} = \alpha_p + \sum_{b=1}^{B} \gamma_k I[BO^2_{cp} = b] + \sum_{t=1}^{T} \tau_t I[t_{cp} = t] + \epsilon_{cp}, \tag{10}$$

   where $\alpha_p = \beta_1 BO^1_p + \nu_p$. Equation 10 corresponds to a sibling fixed effects model. By assumption, the corresponding regression estimates $\hat{\gamma}_k$ and $\hat{\tau}_t$ are consistent for γ_k and τ_t, respectively.

2. Use the estimates from step 1 to construct fitted values $\hat{\gamma}_{cp} = \sum_{b=1}^{B} \hat{\gamma}_k I[BO^2_{cp} = b]$ and $\hat{\tau}_{cp} = \sum_{t=1}^{T} \hat{\tau}_t I[t_{cp} = t]$. Deduct $\hat{\gamma}_{cp}$ and $\hat{\tau}_{cp}$ from both sides of equation 9 such that

$$Y_{cp} - \hat{\tau}_{cp} - \hat{\gamma}_{cp} = \beta_1 BO^1_p + \xi_{cp}, \tag{11}$$

   where $\xi_{cp} = \nu_p + \epsilon_{cp} + \tau_{cp} - \hat{\tau}_{cp} + \gamma_{cp} - \hat{\gamma}_{cp}$. Since BO^1_p is randomly assigned to the parents and is not used in the estimation of $\hat{\gamma}_k$ and $\hat{\tau}_t$, $cov(BO^1_p, \xi_{cp}) = 0$. As a result, a regression of $Y_{cp} - \hat{\tau}_{cp} - \hat{\gamma}_{cp}$ on BO^1_p yields a consistent estimate for β_1.

Regular clustering methods do not yield proper standard errors for the two-step estimator because (i) the number of observations (children) in the sample depends on the treatment assignment and (ii) the first estimation step adds additional uncertainty, and ignoring this will lead to underestimation of the standard errors. Instead, we use a simple clustered bootstrap procedure. Specifically, if there are P families in the sample, then we randomly draw P families with replacement. Next, we apply the two-step estimator to this sample to obtain $\hat{\beta}^1_1$. We repeat this process $R = 200$ times and store the resulting estimates in a vector $\hat{\beta}_{\mathbf{1}} = \{\hat{\beta}^1_1, \hat{\beta}^2_1, ..., \hat{\beta}^R_1\}$. The bootstrapped 95 percent confidence interval for $\hat{\beta}_1$ is then given by the interval between the 2.5^{th} and 97.5^{th} percentile of vector $\hat{\beta}_{\mathbf{1}}$.